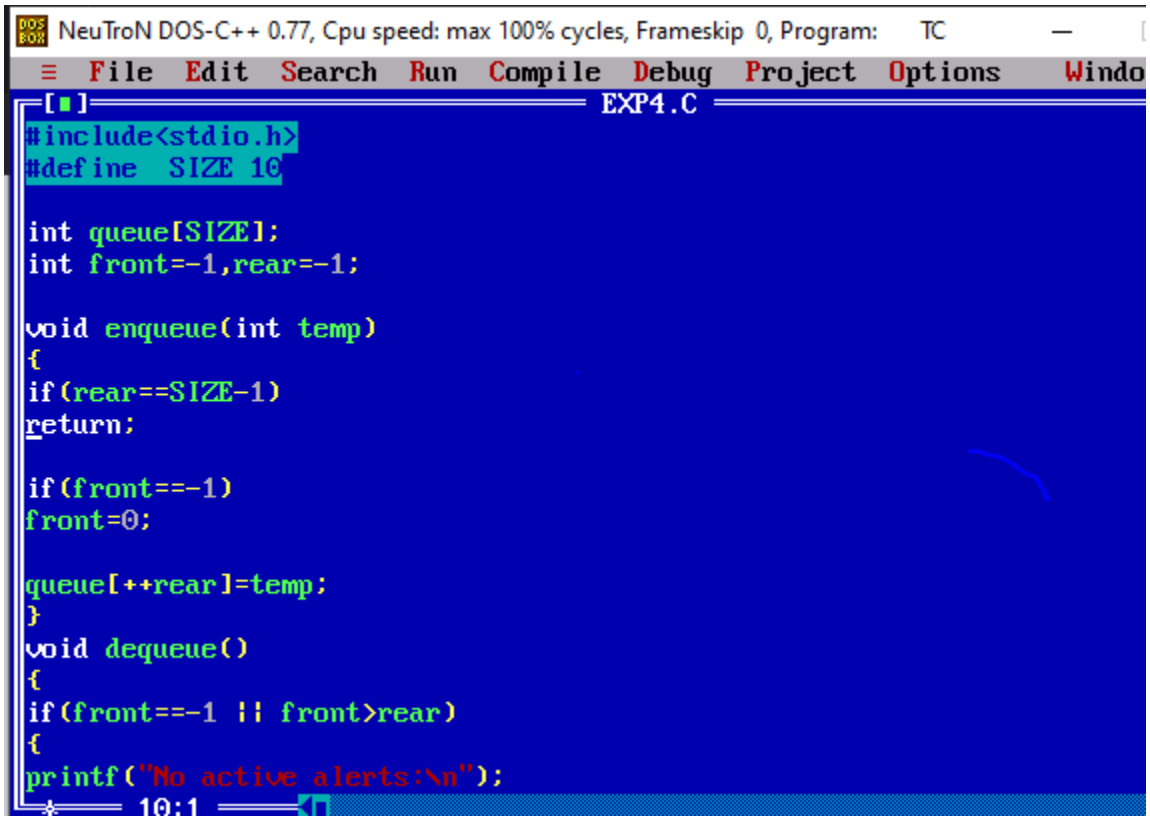
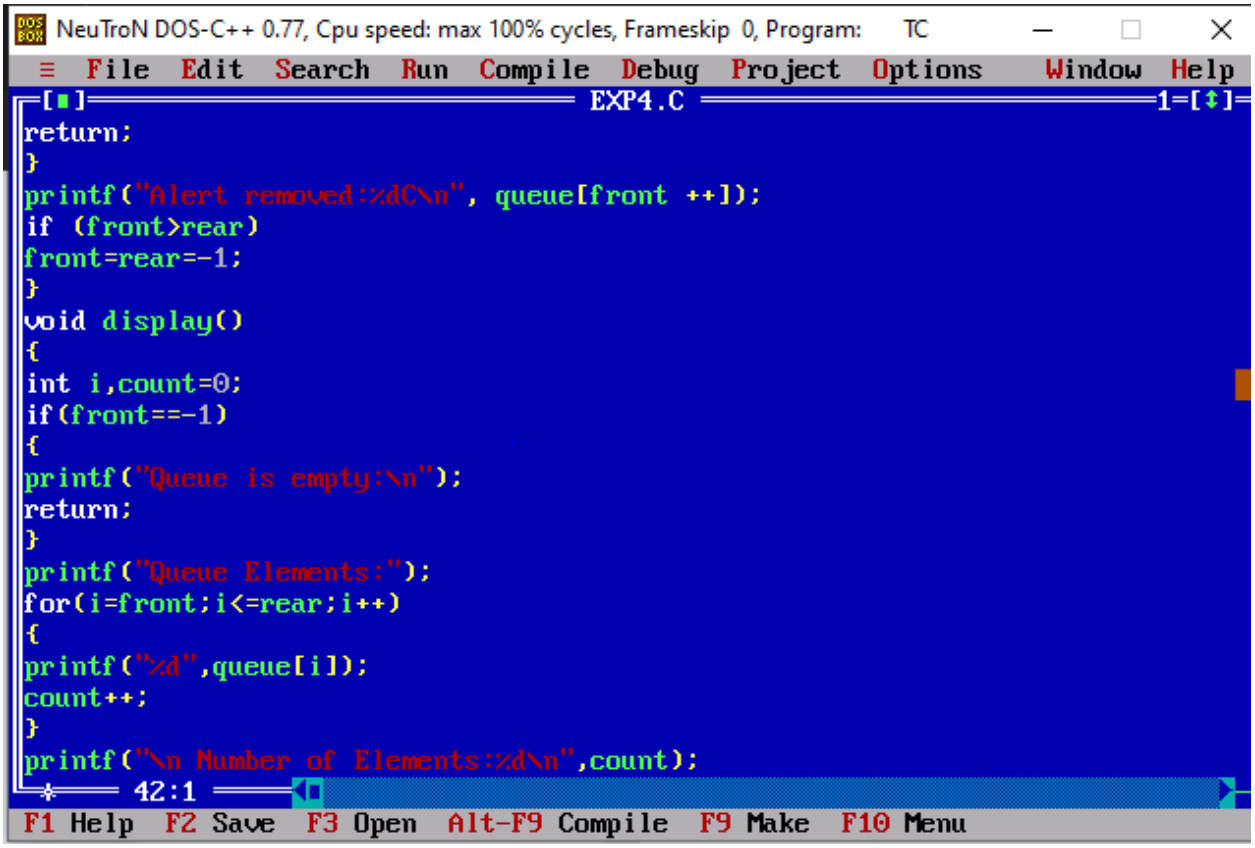
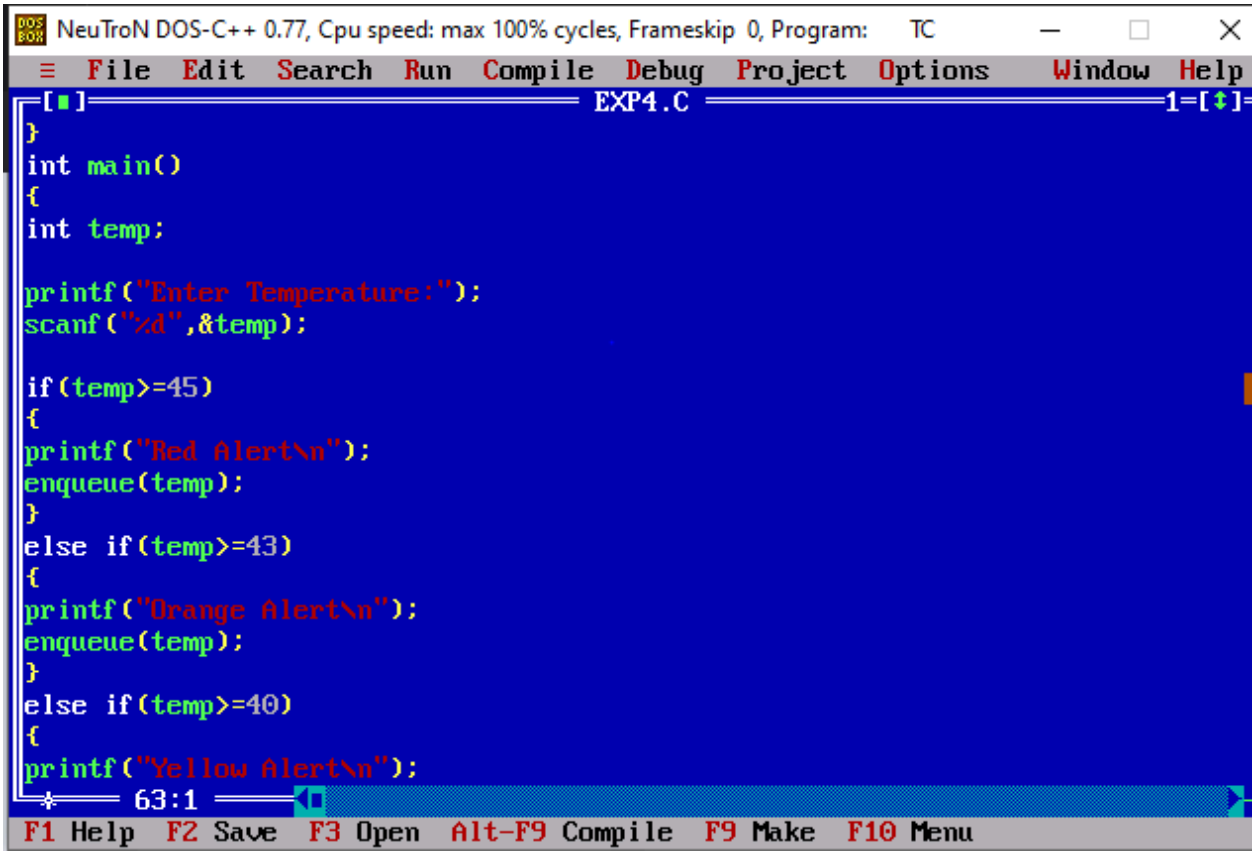
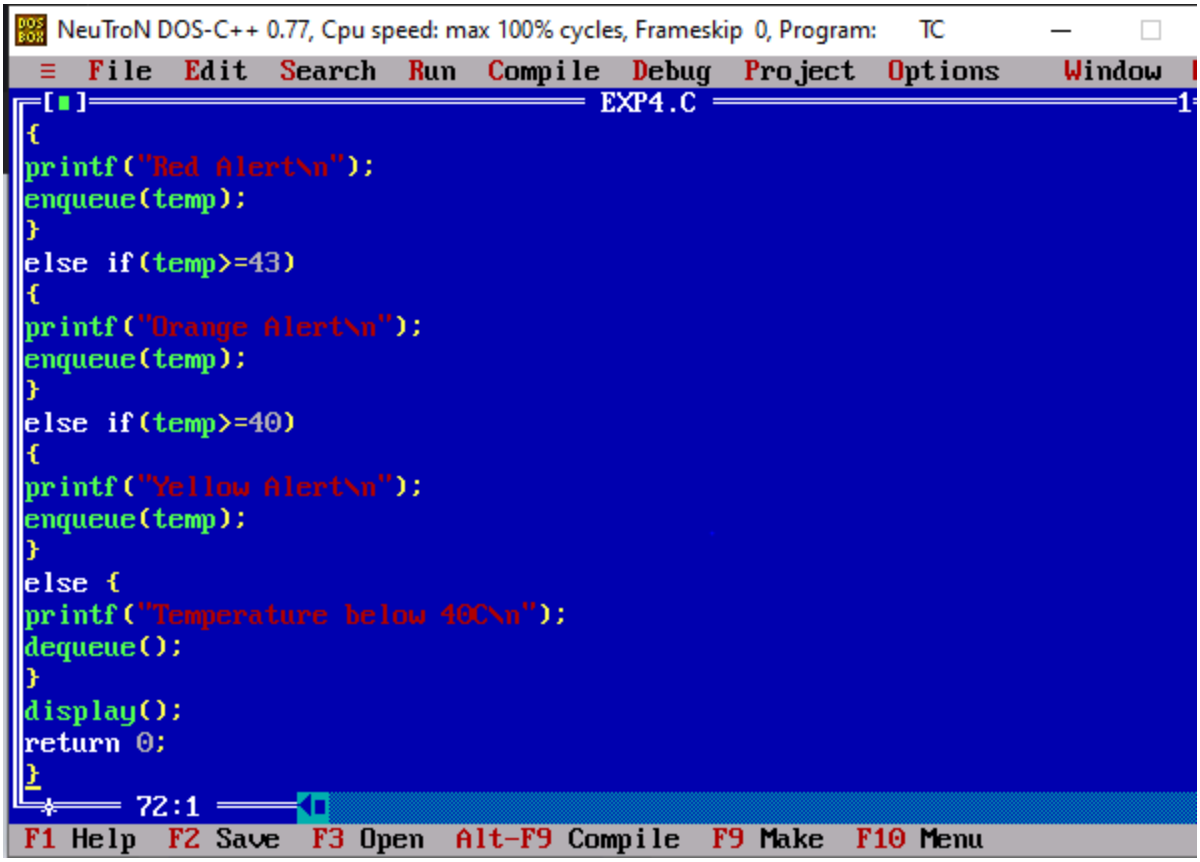
CODE:

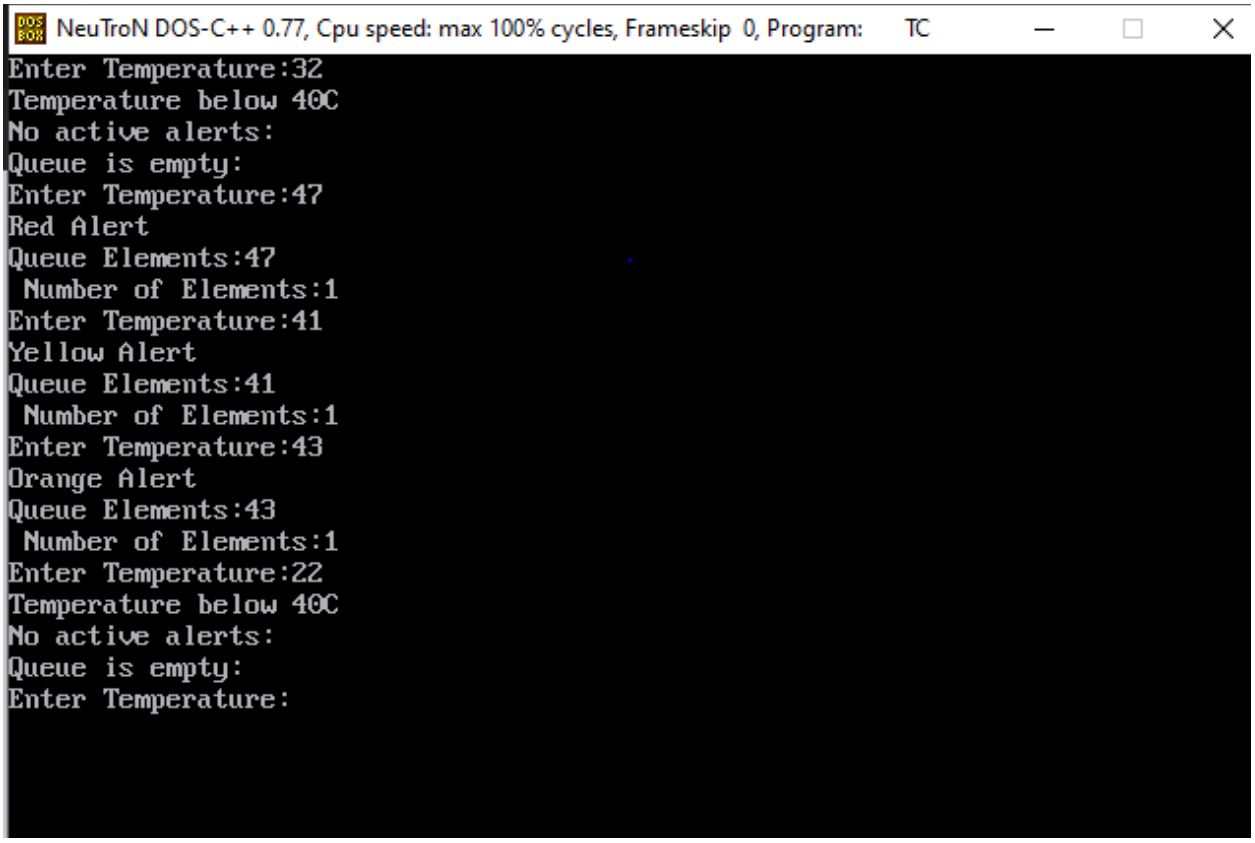

NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Windo
EXP4.C
#include<stdio.h>
#define SIZE 10
int queue[SIZE];
int front=-1,rear=-1;
void enqueue(int temp)
{
if(rear==SIZE-1)
return;
if(front==-1)
front=0;
queue[++rear]=temp;
}
void dequeue()
{
if(front==-1 || front>rear)
{
printf("No active alerts:\n");
10:1


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXP4.C
return;
}
printf("Alert removed:%dC\n", queue[front ++]);
if (front>rear)
front=rear=-1;
}
void display()
{
int i,count=0;
if(front==-1)
{
printf("Queue is empty:\n");
return;
}
printf("Queue Elements:");
for(i=front;i<=rear;i++)
{
printf("%d",queue[i]);
count++;
}
printf("\n Number of Elements:%d\n",count);
42:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXP4.C
}
int main()
{
int temp;
printf("Enter Temperature:");
scanf("%d",&temp);
if(temp>=45)
{
printf("Red Alert\n");
enqueue(temp);
}
else if(temp>=43)
{
printf("Orange Alert\n");
enqueue(temp);
}
else if(temp>=40)
{
printf("Yellow Alert\n");
63:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window
EXP4.C
{
printf("Red Alert\n");
enqueue(temp);
}
else if(temp>=43)
{
printf("Orange Alert\n");
enqueue(temp);
}
else if(temp>=40)
{
printf("Yellow Alert\n");
enqueue(temp);
}
else {
printf("Temperature below 40C\n");
dequeue();
}
display();
return 0;
}
72:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

OUTPUT:


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
Enter Temperature:32
Temperature below 40C
No active alerts:
Queue is empty:
Enter Temperature:47
Red Alert
Queue Elements:47
Number of Elements:1
Enter Temperature:41
Yellow Alert
Queue Elements:41
Number of Elements:1
Enter Temperature:43
Orange Alert
Queue Elements:43
Number of Elements:1
Enter Temperature:22
Temperature below 40C
No active alerts:
Queue is empty:
Enter Temperature: